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Rotary drums with sectional internals: Experimental investigation on the influence of section number and section length

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ABSTRACT

The impact of sectional internals, as often used in industrial rotary drums, on the axial solid transport in rotary drums is investigated experimentally under variation of the operation parameters solid feed rate, drum inclination and rotational speed. Different section numbers and internal-lengths have been analyzed, applying the axial dispersion model, in a fully optically accessible lab scale rotary drum apparatus.
The mean residence times are increasing with the section numbers and their length up to 4 times compared to bare drum results. Capacity limits have been detected, resulting in higher back-spillage ratios dependent on the set of operational parameters. The impacts of parameter changes on the axial dispersion are decreasing with section number and length, resulting in a more constant axial dispersion coefficient of about 1×10^{-6} m^2/s. A qualitative similarity of the axial solid bed profile proposed by the Saeman-model for bare drums has been found for sectional internals.

1. Introduction

Rotary drums are used in solids processing for a long time. Basically, they consist of a cylinder, rotating around its own axis. The axis is slightly inclined to the horizontal and the solid material is introduced at the upper end of the drum. This simple and robust construction leads to the multitude of rotary drum applications from reactive thermic material treatment (ore processing [1], cement production [2,3], pigment production [4,5]) over to non-reactive thermal treatment (drying processes) [6] and other process applications, e.g. mixing and grinding in ball mills [7].

The investigation of the process behavior of rotary drums goes back to the 1920s. Important values for the dimensioning and layout as well as operation are the material hold-up and the residence time (RT), in particular the mean residence time (MRT) and its distribution (RTD). Furthermore, information on capacity limits is very important. The functional dependencies of these values from the geometric dimensions and properties of the drum (length, diameter etc.) as well as from the operational parameters (rotational speed, feed mass flow etc.) and the characteristics of the bulk material (angle of repose etc.) are significant in the design and operation of rotary drums. Basic systematic investigations on these topics have been done by Sullivan et al. in 1927 [8], who gave a first empirical equation to describe the process behavior.

Since then, many research groups have investigated the rotary drum. Empirical models, based on dimensionless numbers, have been developed by Chatterjee [9,10] to calculate the MRT ($\bar{t}$) as one important value in the description of axial solid transport in bare drums:

$$\bar{t} = 0.1026 \cdot \frac{L^3}{\dot{V}} \cdot \left(\frac{\theta}{\beta}\right)^{1.054} \cdot \left(\frac{\dot{V}}{L^3 n}\right)^{0.981} \cdot \left(\frac{L}{D}\right)^{1.1} \tag{1}$$

One of the best known models, which is widely used in dimensioning and layout until today, is the Saeman-model from 1951 [11]. It is based on a geometrical approach and leads to a first order differential equation (ODE), describing the solid bed height (h) profile in the axial direction (x) dependent on geometrics of the bare drum and operational parameters [11–13]:

Abbreviations: EC, Equipment Configuration; Exp., Experiment; IF_{EC}, Increase factor regarding equipment configuration; IF_L, Increase factor regarding section lengths; MIF, Mean increase factor; MRT, Mean Residence Time; ODE, Ordinary Differential Equation; PMMA, Polymethylmethacrylate; RT, Residence Time; RTD, Residence Time Distribution.

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$$\frac{dh}{dx} = \frac{3 \cdot \dot{V}}{4 \cdot \pi \cdot n} \cdot \frac{\tan(\theta)}{\left(2Rh - h^2\right)^{\frac{3}{2}}} - \frac{\tan(\beta)}{\cos(\theta)} \tag{2}$$

Similar relations as description of the solid bed even for completely horizontal drums have been found by others [14,15]. Integrating Eq. (2) and the related cross section results in a mathematical expression for the hold-up volume V_{hold}. The MRT can hereby be calculated from this hold-up and the volumetric flow-rate $\dot{V}$ to the hydrodynamic residence time τ_{hyd}:

$$\tau_{hyd} = \frac{V_{hold}}{\dot{V}} = \frac{m_{hold}}{\dot{m}} \tag{3}$$

Equally, the calculation can be performed with the hold-up mass (m_{hold}) and the mass flow rate ($\dot{m}$). The approach is used until today to model the solid bed in rotary drums e.g. by Descoins et al. and Ndiaye et al. [16,17]. Although it was originally developed for bare rotary drums, it has also been used to model the behavior of rotary drums with diverse constriction dams e.g. by Kramers et al. and Scott et al. [14,18].

However, the research on rotary drums is not only restricted to the investigation of the process behavior of bare drums, since several internals can be used e.g. lifters, constriction dams, chains etc. A good overview of the used internals in rotary kilns is given from Duda and Boateng [2,3]. In detail, especially the design of rotary drums is therefore very individual and versatile. The impact of lifters or flights has been investigated intensively by e.g. Sunkara et al., Sheehan et al. and others [19–22]. Abouzeid et al. and Chatterjee et al. investigated the influence of constriction dams within the drum or at the discharge end of the drum [10,23].

This study concentrates on rotary drums with sectional internals as often described and used in industrial applications e.g. as heat exchanger internals [3,24–26]. The cross section of the rotary drum is divided into sections with walls running radially from the center of the drum to the cylinder wall. In industry, these sectional internals are built out of steel or ceramic material depending on the temperature regime in which they are installed.

A first orienting analysis has been conducted by our working group, presenting a massive influence of sectional internals on the process behavior [27]. Additionally, the impact of sectional internals has been analyzed in the cross-section of the drum [28]. However, the scientific investigation and description in this field is unfortunately insufficient so far, therefore, the impact of these sectional internals in different configurations is investigated in this study experimentally. Besides the operational parameters, different numbers of sections and lengths of the sectional internals have been varied.

2. Experimental setup/methodology

A completely new, innovative and optically accessible experimental setup, which was developed to measure residence time distributions (RTD) flexibly regarding the internal structures of the drum, is the basis for this study.

2.1. Rotary drum apparatus

Two main aspects determined the design and construction of the novel laboratory rotary drum device. Since many drum configurations are investigated, flexibility in mounting different possible internals is essential to minimize effort and maximize comparability of the results as first point. Secondly, in order to observe and measure solid transport inside the drum non-invasively, full optical access of the model enables the possibility of particle flow analysis via optical techniques, analogously to other equipment, e.g. spiral jet mills [29,30].

Additionally, the ability to measure the influence of sectional internals at different operational states is essential, which demands variability concerning process parameters as rotational speed, feed rate and drum inclination. Fig. 1 presents the used apparatus as 3D-model (right) and in a picture of the laboratory arrangement (left).

All equipment is mounted on a base frame. A screw conveyor (Fig. 1, Pos. 1; Gericke GLD75, 50 W) delivers the desired feed mass flow via a frequency-controlled motor. A hopper on top of the screw conveyor serves as bulk storage. A funnel and a chute (Fig. 1, Pos. 2) conduct the feed mass flow into the inlet end of the rotary drum. The chute is induced into the inlet end of the drum via a sealing housing. The housing

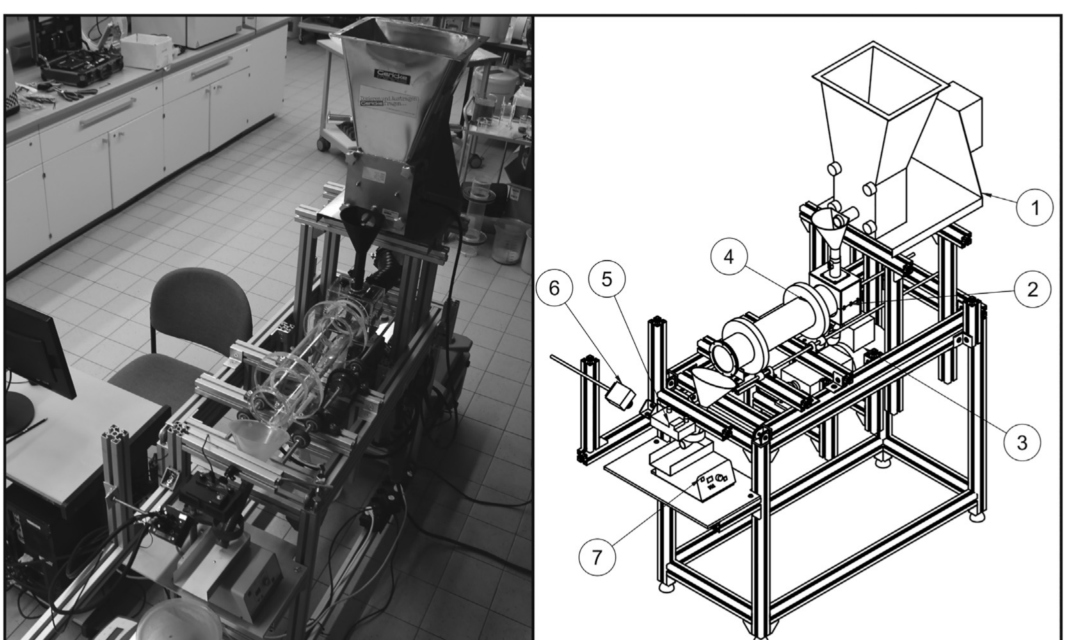

Fig. 1. Rotary drum apparatus used in this study. Left: Photo of the laboratory installation; Right: Sketch of the apparatus with important parts.

has another funnel with a connected vessel at the bottom to collect material, possibly coming out of the inlet end of the drum (back spillage).

The drum (Fig. 1, Pos. 4) itself is constructed out of a PMMA tube with two PMMA tires (transparent, optically accessible). It is mounted on a roller station consisting of two axes with four rollers. In order to prevent the axial drum drifting due to inclination, a tiny horizontal roller is running to one of the drum's tires. Different rotational speeds can be set to the rotary drum by a motor (Fig. 1, Pos. 3; Heynau H-Trieb, 250 W) with integrated variable mechanical transmission. One axis of the roller station is equipped with a sprocket connected to the power unit by a chain drive.

At the outlet end the solid mass flow is collected with a funnel deducing it onto a vibration conveyor (Fig. 1, Pos. 7; Retsch DR100) with modified chute afterwards. The solids on the vibration conveyor are illuminated from the top by LED panels (Fig. 1, Pos. 5). A camera (Fig. 1, Pos. 6, Vieworks VC-2MC-C340EO) observes the whole cross section of the chute and thus the solid material on it. A computer is used to trigger the camera via a frame grabber (Xtium-CL_MX4_1) and as storage device.

The equipment is grounded at all possible positions to prevent electrostatic charging via wire connections to the base frame. The base frame itself is connected to the ground.

The rotary drum itself is equipped with two fix flanges at both ends, used to easily mount and exchange any internals or extensions. The sectional internals are constructed as slide-in modules and fixed to the drum's ends with flanges. The sectional internals are also constructed out of PMMA to guarantee optical accessibility again. No further treatment or modification of the wall surfaces has been applied. The thickness of the material used to build the segmentation is 5 mm. In this study, seven different drum configurations were investigated, namely three different sectional internal structures, each with two lengths and the bare drum. The cross sections of the configurations are shown in Fig. 2.

The first equipment-configuration (Fig. 2, EC1) represents the bare drum without any internals or constriction dams at the outlet end. In the second equipment-configuration (Fig. 2, EC2) sectional internals are used (analogously to the description before) dividing the whole cross section into three equal “pie slices” or compartments. The third and fourth configuration (Fig. 2, EC3 and EC4) are similar to EC2, but the cross section is now divided into four or six identical sections respectively.

The lengths of the sectional internals were varied in two stages. A shorter version (L1) is 225 mm long measured in axial direction from the outlet end of the drum. The longer version (L2) is 450 mm long, because the chute is reaching into the drum at the inlet end and limiting the length. The lengths of the internals are presented in Fig. 3 together with the detailed geometric structure of the drum.

The bar in the middle of the drum in Fig. 3 symbolizes one spoke of the sectional internals. The total length of the tube is 510 mm, the inner diameter is 100 mm. The L/D ratio thus has a value of 5. The inlet end is equipped with a constriction dam with inner diameter of 60 mm to prevent/reduce back spillage of the particle feed. This inlet constriction dam is used for all configurations in this study. At the discharge end, no restriction dams are applied in this study.

As nomenclature, a combination from the cross sectional equipment configuration (EC) and the length (L) with ECxLy is used in this examination to describe the used drum configuration. The bare drum is always called EC1 because there is solely the drums' length of 510 mm.

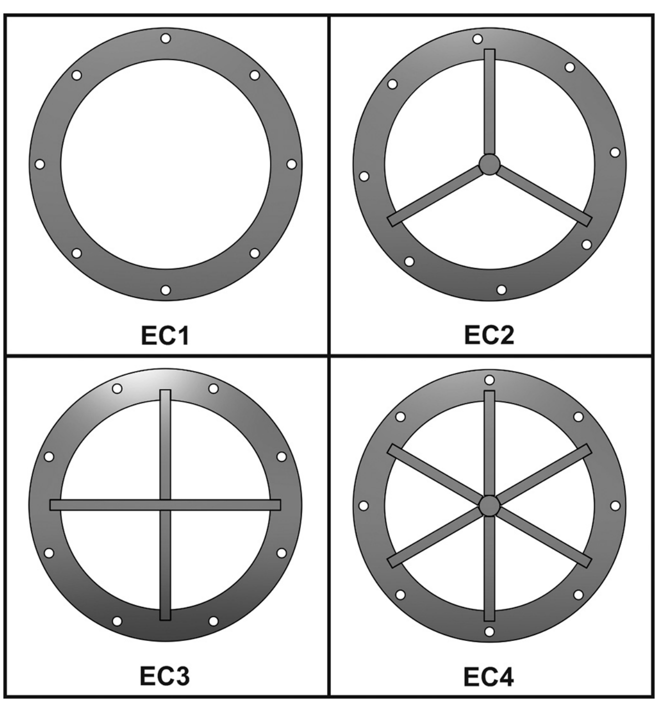

Fig. 2. Sketch of cross sections of the used drum configurations (equipment configurations, EC). EC1: bare drum; EC2: 3 sections; EC3: 4 sections; EC4: 6 sections.

2.2. Solid material

In order to prevent influences as e.g. segregation-effects due to a wide particle size distribution or density-effects evoked by a gangue etc. [31], glass particles were used as bulk material in this study. They have a spherical shape, the solid density of all particles is the same and the particle size distribution is narrow. Furthermore, glass particles have a uniform color in contrast to natural products, as e.g. sand.

Since the analyzing-technique is based on different colors, black-colored glass particles are used as tracer. The tracer is manufactured from the bulk material with water-soluble dye to ensure color as the only difference between tracer and bulk particles. The used material is collected after each experiment and washed, dried and sieved afterwards. Although there are no different particle characteristics except color, full recycling of the material is enabled by this technique. A summary of particle characteristics is given in Table 1.

Particle characteristics as particle size, sphericity and solid density are known from the product data sheet of the manufacturer (Sigmund

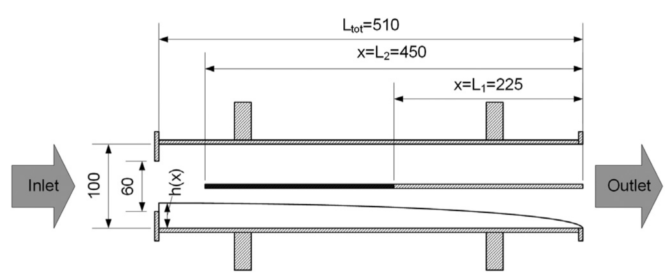

Fig. 3. Side view-sketch of the drum with all dimensions and different lengths of internals, all values in mm.

Table 1
Characteristics of used bulk materials.

Property		Bulk (white)	Tracer (black)
Particle size	μm	100–200	100–200
Sphericity	–	≥ 0.89	≥ 0.89
Bulk density	kg/L	1.454 ± 0.002	1.331 ± 0.002
Solid density	kg/L	2.50	2.50
stat. angle of repose	°	18.9 ± 0.7	23.6 ± 1.0
dyn. angle of repose (Θ)	°	30.0 ± 1.0	33.0 ± 1.0

Lindner GmbH). Stability of the particle size is gained via sieving before each experiment. Additionally, the static angle of repose (H-W-S Labortechnik, DIN ISO 4324) and the bulk density (HLL Landgraf, DIN ISO 697) were measured. The dynamic angle of repose is measured with the cross section apparatus described in [28], having the same diameter (0.1 m) and material (PMMA) as the apparatus in this study. The given value is a mean value from tests with different rotational speeds in the range of the DOE used in this study.

2.3. Experimental procedure and parameter settings

The experiments were conducted referring to the following procedure:

1. The chosen parameters for rotational speed, drum slope and mass-flow rate are set and checked. The hopper of the screw conveyor is filled with a sufficient amount of solid material to serve the experiment.
2. Steady-state conditions are set by measuring the mass-flow rate at the outlet until no further drift occurs (differences <3%). A thin material film is generated on the vibration conveyor at the outlet end via frequency adjustment.
3. The drum and the feed screw are stopped and the collecting vessels behind the inlet and outlet end of the drum are emptied from the material needed for steady state setting. The weight of the vessels is determined, and the vessels are repositioned at the drums ends.
4. The apparatus is started again, dyed tracer is injected (25 g) in <1 s, which is very short compared to measured residence times and the picture acquisition is started simultaneously. Thus, the tracer injection is considered as pulse function (Dirac impulse).
5. The whole equipment and the experiment are stopped when the tracer completely left the rotary drum with a safety time. The absence of tracer is checked by visual inspection via operators´ eyes and camera observation. Additionally, the times have been determined in preliminary experiments not part of this study.
6. The masses of the collecting vessels and therefore the weights of the material at the outlet and inlet end of the drum are measured.
7. The solid material inside the rotary drum is discharged and its weight is measured as hold-up.
8. The photo sequence of the outlet flow on the vibration conveyor is analyzed as illustrated in 2.4.
9. The solid material is prepared for the next measurement by washing, drying and sieving.

Different combinations of the three operational parameters are investigated based on a statistical or orthogonal design of experiments. Each of the seven configurations was analyzed with the same experimental design, in total this results in 63 experiments.

Table 2 presents the chosen setting combinations for each drum configuration.

Table 2
Design of experiments (full-factorized design, three factors on two stages) for the chosen parameter combinations in the experiments.

Exp.-No.	Rotational speed	Slope	Feed rate
	n [1/min]	β [°]	$\dot{m}_F$ [kg/h]
01	3.8	1.3	9.5
02	11.0	1.3	9.5
03	3.8	4.3	9.5
04	11.0	4.3	9.5
05	3.8	1.3	16.0
06	11.0	1.3	16.0
07	3.8	4.3	16.0
08	11.0	4.3	16.0
09	7.3	3.0	13.5

2.4. Analyzing method

The experimental procedure with the described apparatus uses a contactless analyzing method without a need for invasive sampling. The data acquisition works as an inline technique (picture acquisition) while further data processing is done afterwards in a post-processing step using digital image processing techniques.

1. Data acquisition

The high-speed camera takes photos of the vibration conveyor surface with particles in defined, triggered time intervals. The analyzing rate (2 pictures/s) is high compared to the measured residence times and keeps handling of the data amount feasible. The exposure times are low (5 ms), due to the observation of moving objects/particles. In order to guarantee the same possibility and probability for each particle to be visible on the picture, the whole cross section of the chute (100 mm) is evaluated. Since no particle should be analyzed twice, the observation area is small in flow direction (7 mm).

The bulk glass particles are nearly invisible, since the LED panels and the surface-preparation of the vibrating chute in white color illuminate them. Consequently, a very high contrast to black tracer particles is generated to improve sensitivity of the technique.

The result of the measurement is a picture sequence of the particles or the particle flow at the end of each experiment.

2. Data processing

The software ImageJ is used to perform the evaluation of the picture sequence via digital image analysis. In a first step, the first picture of the sequence is subtracted from each following picture as blind value. The ratio of black area to the analyzed area is proportional to the number of tracer particles in each photo. In addition, the number of tracer particles is proportional to their concentration. The black area is measured with a defined threshold for all pictures in the whole picture sequence. The experimental time is connected to the picture-numbers via the camera setup (trigger intervals). In conclusion, plotting the black area fraction against time directly leads to the residence time distribution.

2.5. Examination of data

Characteristic values are calculated from the measured data (residence time distribution) to provide a better comparison of results. The axial dispersion model is used to fit the measured residence time distribution (RTD). The axial dispersion model is based on convective material transport superimposed by dispersive/diffusional material transport [32–34]:

$$\frac{\partial c}{\partial t} = D_x \cdot \frac{\partial^2 c}{\partial x^2} - v_x \cdot \frac{\partial c}{\partial x} \tag{4}$$

The partial differential equation describes the concentration c, which is changing with the time t at an axial position x. A dimensionless expression of Eq. (4) is used to determine the coefficients v_x (velocity in the observed axial dimension) and D_x (dispersion coefficient in the observed axial dimension) [35]:

$$\frac{\partial C}{\partial \vartheta} = \left(\frac{D_x}{v_x L}\right) \cdot \frac{\partial^2 C}{\partial z^2} - \frac{\partial C}{\partial z} = \frac{1}{Pe} \cdot \frac{\partial^2 C}{\partial z^2} - \frac{\partial C}{\partial z} \tag{5}$$

The used expressions are the dimensionless concentration C, dimensionless time ϑ and the dimensionless length z, respectively. The dimensionless group in Eq. (5), called as Peclet-number (Pe), represents the relation of dispersive motion and convective flow:

$$Pe = \frac{v_x L}{D_x} \tag{6}$$

Several analytical solutions with different boundary conditions of the partial differential Eq. (5) have been provided, e.g. by Nauman [33]. The solution of Eq. (5) for open-open boundary conditions [34,35] is:

$$E_\vartheta = \bar{t} \cdot E = \frac{1}{\sqrt{4\pi Pe}} \cdot \exp\left[-\frac{(1-\vartheta)^2}{4Pe}\right] \quad (7)$$

Eq. (7) considers no disturbances in flow at the inlet and outlet end and small extends of dispersion (Pe > 100). This leads to a symmetrical shape of the curves for the density distribution (E) and the similarity to the Gaussian distribution is obvious. The Gaussian distribution with the parameter residence time t on the abscissa is presented in Eq. (8):

$$E = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left[-\frac{(t-\bar{t})^2}{2\sigma^2}\right] \quad (8)$$

For further determination, the Gaussian distribution is fitted to the measured RTD-curves (E) via least squares-method. Mean residence time ($\bar{t}$, maximum of distribution) and the standard deviation or variance σ are the used fit factors. A comparison of coefficients in Eqs. (7) and (8) delivers the following relation to calculate the Peclet-number via the fitted Gaussian distribution [35,36]:

$$\sigma_\vartheta^2 = \frac{\sigma_t^2}{\bar{t}^2} = \frac{2}{Pe} = 2 \cdot \left(\frac{D_x}{v_x L}\right) \quad (9)$$

With the drum length L and the mean residence time $\bar{t}$, the mean axial velocity v_x is determined to:

$$\bar{v}_x = \frac{L}{\bar{t}} \quad (10)$$

The axial velocity and the Peclet-number are used according to Eq. (9) to calculate the axial dispersion coefficient and thus the amount of axial back-mixing in the drum.

The differential Eq. (4) can now be used with the two parameters v_x and D_{ax} of the axial dispersion model to describe the solid transport through the rotary drum (e.g. in modeling).

3. Results and discussion

The experiments provide novel results, which are presented in the following chapter. Besides quantitatively measured values, the optical accessibility is used to deliver exploring views inside the drum.

3.1. Quantitative results

In this study, no multiple measurements were carried out in order to keep the high experimental effort due to the parameter and configuration variation feasible. A good reproducibility especially for the mean residence time (MRT) and hold-up values has been shown with a comparable apparatus and methodology [27]. Rolling bed motion was observed qualitatively in the bare drum parts by visual investigation in all experiments. Table 3 shows all the measured data (determined according to chapter 2.5). The values for EC1 are assigned to the length L2 since L2 with 450 mm is closer to the length of the drum (510 mm).

3.1.1. Mean residence times

The experimental MRT-results are presented graphically in Fig. 4 in order to analyze trends and dependencies visually. Since the variation of the operational parameters already covers three dimensions and additionally the drum configuration has been changed, a multi-dimensional plot is not possible.

Therefore, Fig. 4 presents the results of the MRT in two stacked bar charts. The colors refer to the equipment configuration or number of sections, respectively. The upper diagram exhibits the results of the shorter sectional internals (L1), the lower diagram displays the results of the longer sectional internals (L2, indicated with hatchings). From this representation, a multitude of trends can be evaluated:

- In general, the MRT is dependent on the combination of operational parameters set for each experiment.
- In direct comparison to the EC1-results, MRT-values calculated by the models of Chatterjee and Saeman (cf. chapter 1) are displayed. A reasonable congruence is obvious, except some outliers. This congruence is a good validation of the measurements and the dependencies due to operational parameters and literature models. Higher deviations have been found at Exp. 01/05 with the Saeman-model and at Exp. 03/07 with the Chatterjee-model. The deviation of the Saeman-model is evoked by the occurrence of back-spillage, as explained later.
- An increase of the feed mass flow has only very little effect on

Table 3
Results of mean residence time (MRT), hold-up and axial dispersion coefficient (D_{ax}) for all measurements.

	Exp.-	MRT [s]				Hold-Up [g]				D_{ax} [x10^{-6} m^2/s]			
	No.	EC1	EC2	EC3	EC4	EC1	EC2	EC3	EC4	EC1	EC2	EC3	EC4
L1	1		503	669	1008		894.6	960.2	1038.7		0.73	0.84	0.78
	2		214	264	384		617.2	759.9	972.5		2.18	1.65	0.82
	3		202	239	337		653.2	805.2	975.0		1.55	1.10	1.13
	4		79	89	122		214.7	244.2	335.2		7.89	6.36	4.02
	5		490	660	991		967.7	1036.0	1098.9		0.81	1.00	0.67
	6		195	247	374		850.1	934.8	1034.9		1.59	0.98	0.35
	7		209	254	375		988.5	1169.0	1461.3		1.80	2.04	1.02
	8		78	90	123		350.7	404.8	566.6		5.14	4.64	3.60
	9		162	186	260		661.2	749.3	1056.8		2.50	1.61	1.64
L2	1	320	740	940	1372	679.1	920.3	986.4	1006.5	0.68	1.71	1.31	0.19
	2	146	282	368	507	420.4	811.6	910.3	986.9	3.43	0.93	0.24	1.02
	3	147	271	355	522	418.2	789.4	1024.8	1272.2	2.45	0.57	0.84	1.14
	4	58	102	129	179	155.6	277.4	365.7	488.8	13.18	2.69	1.56	1.30
	5	311	736	927	1312	729.5	990.8	1053.7	1071.4	0.62	3.40	2.03	0.76
	6	129	268	364	493	595.6	902.8	968.4	1041.0	4.30	0.94	0.63	0.90
	7	145	281	368	524	663.5	1165.6	1296.1	1349.7	2.54	1.26	0.95	0.94
	8	57	100	129	181	253.9	452.5	589.1	838.1	10.28	2.30	1.73	1.19
	9	107	208	262	369	424.2	884.2	1044.9	1174.6	3.72	2.05	0.72	0.94

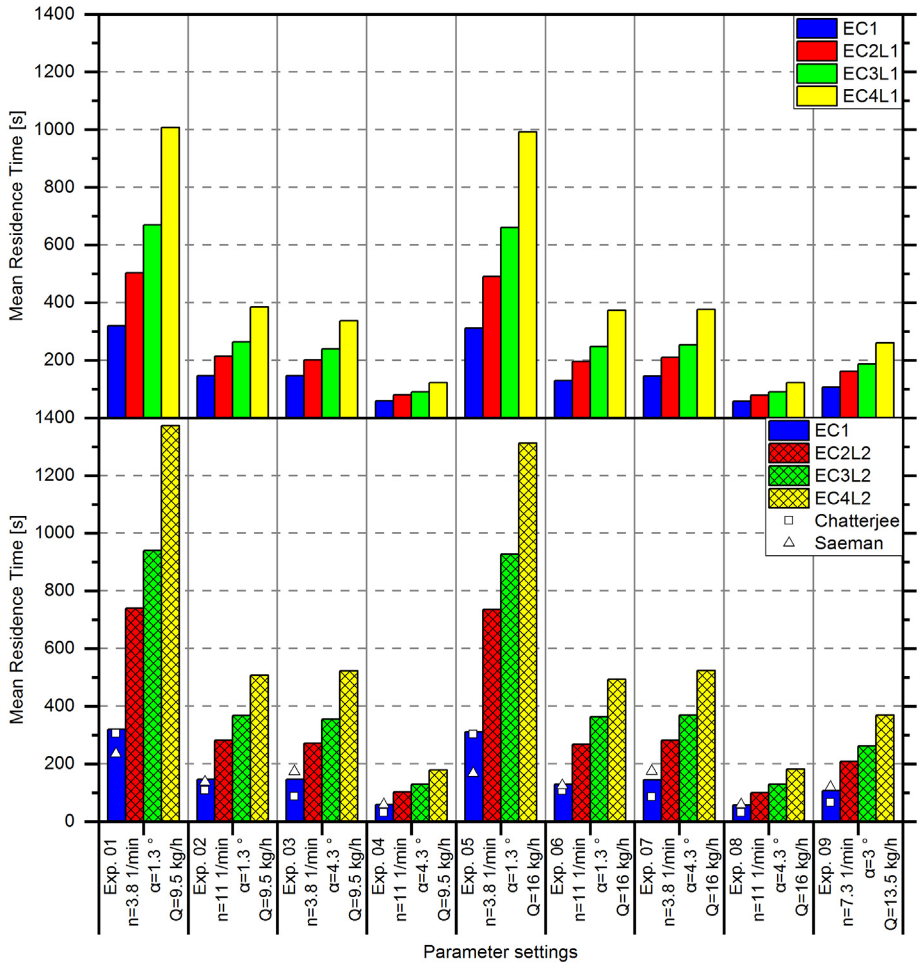

Fig. 4. Comparative diagram of the MRT-results for all drum-configurations. Upper diagram: L1 = 225 mm section lengths; lower diagram: L2 = 450 mm section lengths.

residence time when all other parameters are kept constant (cf. Exp 01/05; Exp. 02/06 etc. for all configurations).

- The rotational speed of the drum has a very large impact on the residence time (cf. Exp. 01/02; Exp. 03/04 etc. for all configurations). A higher rotational speed leads to significant shorter MRT.
- The inclination angle of the drum also has a very large effect on the MRT (cf. Exp. 01/03; Exp. 02/04 etc. for all configurations). Significantly lower MRTs have been measured with higher inclination angles of the drum.
- The observed trends are also represented in the effect-analysis (chapter 3.1.5) and explained in detail there.

The above trends concerning the three operational parameters are valid for the bare drum and all internal configurations of the drum. They are also valid for both lengths of the sectional internals. As additional novelty, trends due to the number of sections and the lengths of the sectional internals are obvious.

Since the dependencies regarding the operational parameters are similar, an analysis of increase factors is conducted. The factors between MRT of the bare drum and the MRT with different configurations (IF_{EC}) are evaluated for each experiment. Furthermore, a comparison of the longer and the shorter version is performed (IF_L). The results are presented in Table 4.

Additionally to the visual findings in Fig. 4, it is clearly observable from the mean increase factors (MIF) in Table 4, that:

- The number of sections is influencing the MRT. A higher section number leads to an increase in MRT. This is valid regardless of the internal lengths.
- The length of the sectional internals additionally impacts the MRT due to longer MRT with higher lengths of the sectional internals. However, the MIF_L-results with factors of roughly 1.4 show, that doubling the lengths of sectional internals (L2 = 2*L1) does not lead to a doubling in MRT.

The mean factors are presented graphically in Fig. 5. A factor of 1 is set for the bare drum with an assumption of 1 as section number. Both lengths give an excellent linear correlation (R^2 above 0.95),

Table 4
Results of increase factor analysis (IF_{EC}) for MRT between different drum configurations and between different internal lengths (IF_L).

	IF_{EC}						IF_L		
	L1			L2					
No.	EC2	EC3	EC4	EC2	EC3	EC4	EC2	EC3	EC4
1	1.57	2.09	3.15	2.32	2.94	4.29	1.47	1.40	1.36
2	1.46	1.80	2.62	1.93	2.51	3.46	1.32	1.40	1.32
3	1.38	1.63	2.30	1.85	2.42	3.56	1.35	1.48	1.55
4	1.36	1.53	2.10	1.75	2.21	3.07	1.29	1.44	1.46
5	1.58	2.13	3.19	2.37	2.99	4.23	1.50	1.40	1.32
6	1.51	1.92	2.89	2.07	2.82	3.82	1.37	1.47	1.32
7	1.45	1.75	2.60	1.95	2.55	3.63	1.35	1.45	1.40
8	1.38	1.59	2.16	1.76	2.26	3.19	1.28	1.42	1.48
9	1.52	1.75	2.44	1.95	2.46	3.47	1.28	1.40	1.42
Mean	1.47	1.80	2.61	1.99	2.57	3.64	1.36	1.43	1.40
rel. Std.	0.06	0.12	0.16	0.11	0.11	0.12	0.06	0.02	0.06

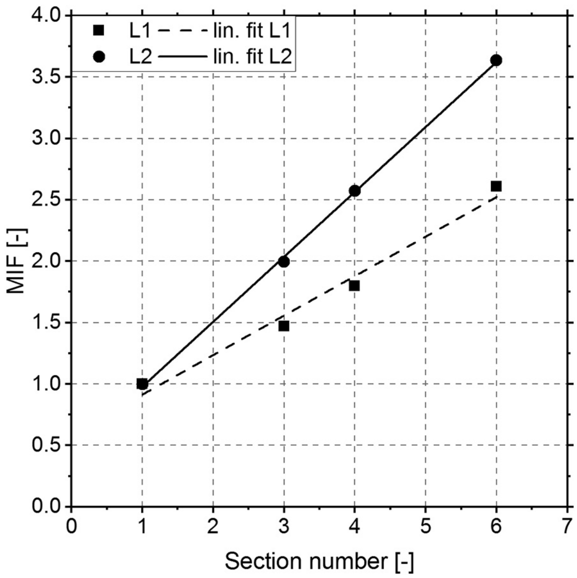

Fig. 5. Mean increase factors and linear fits (R^2 above 0.95). Slope for L1-fit is 0.3 and for L2-fit 0.5 respectively.

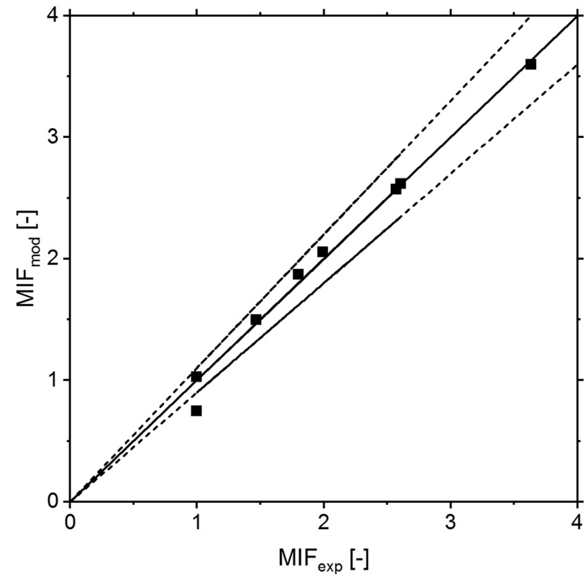

Fig. 6. Diagram comparing the experimental MIF-values and the modeled MIF-values in the modeling of MRT impact due to sectional internals. Dashed lines indicate ±10% deviation margins.

indicating a linear increase of the MIF with section number. The slope regarding long internals (L2) is 0.5 and for short internals (L1) 0.3 respectively.

The slope itself is therefore dependent on the internal length. In order to evaluate a function describing the MRT-behavior of rotary drums with sectional internals, an approach of a family of linear functions according to the following equation is chosen:

$$MIF_{mod} = a\lambda^b \cdot N + a\lambda^b \tag{11}$$

The factor λ is the dimensionless fraction of drum length occupied by sectional internals ($\lambda_1 = 0.45$; $\lambda_2 = 0.9$). This factor is fitted in a polynomial approach with two fit parameters (a, b) via least squares method to the experimentally determined MIF-values. The function describing MIF finally is:

$$MIF_{mod} = 0.54\lambda^{0.46} \cdot N + 0.54\lambda^{0.46} \tag{12}$$

A very good congruence is achieved as presented in the diagram in Fig. 6, comparing experimental and modeled MIF-values. There is only one outlier leaving the 10% deviation margins, namely the modeled factor describing the pseudo bare drum at L1. However, although the model equation is dimensionless, its derivation is empirical and based on the absolute drum dimensions used in this study (D = 0.1 m and L = 0.5 m), as well as section numbers $N = 3/4/6$ and sectional internal occupancies (λ=0.45/0.9). Thus, the strict validity is limited to these parameter ranges in a first step and the scale-up or general validity of the model has to be checked in further measurements with different drum dimensions.

Concluding, based on the experimental findings, a calculation approach based on an increase factor according to the bare drum-MRT is developed. For industrial practice and design calculations, the MRT for bare drums according to the operational parameters can be determined according to existing models (e.g. Saeman or Chatterjee). Afterwards, it has to be manipulated with the MIF according to the drum configuration.

3.1.2. Back-spillage

For some experiments or parameter settings, the system shows an overrun. That means, solid material is unwantedly coming out of the drum on the inlet end, which is called back-spillage. The amount of back-spillage (mass flow $\dot{m}_B$) measured in the experiments is calculated as fraction f_B of the feed mass flow rate:

$$f_B = \frac{\dot{m}_B}{\dot{m}_F} \tag{13}$$

The results are presented in Table 5.

A comparison of the data also reveals a pattern. Similar to the residence time, back-spillage is also dependent on the operational parameters. With higher feed mass flow rates or lower inclination angles of the drum the risk of back-spillage increases massively. If it occurs, the amount of back spillage is increasing in the same manner. These observations have also been found by Liu et al. and Chatterjee et al. for bare drums or drums with constriction dams [13,37]. Back spillage thus behaves antiproportionally to the MRT, which is plausible presupposing similar hold-ups in the drum, as will be discussed later.

Besides dependencies on the operational parameters, the mean values of f_B reveal clear trends regarding the equipment configuration (derivable from mean values in Table 5). The standard deviations are high, since absolute f_B-values occupy a great range due to the different operational parameters from the DOE. However, the mean values are consistent with the characteristic development of back spillage regarding the EC for each set of parameters. A higher number of sections or an

Table 5
Back spillage-results for all measurements as fraction of the feed mass flow rate f_B.

		L1			L2		
Exp.-No	EC1	EC2	EC3	EC4	EC2	EC3	EC4
1	0.33	0.39	0.51	0.66	0.59	0.64	0.76
2	0.00	0.00	0.00	0.15	0.03	0.16	0.34
3	0.00	0.00	0.00	0.00	0.00	0.00	0.18
4	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	0.50	0.59	0.67	0.78	0.71	0.77	0.84
6	0.00	0.08	0.21	0.43	0.32	0.39	0.56
7	0.00	0.00	0.01	0.19	0.12	0.27	0.48
8	0.00	0.00	0.00	0.00	0.00	0.00	0.00
9	0.00	0.00	0.00	0.00	0.00	0.01	0.24
Mean	0.09	0.12	0.16	0.25	0.20	0.25	0.38
Std.	0.19	0.22	0.26	0.30	0.28	0.29	0.31

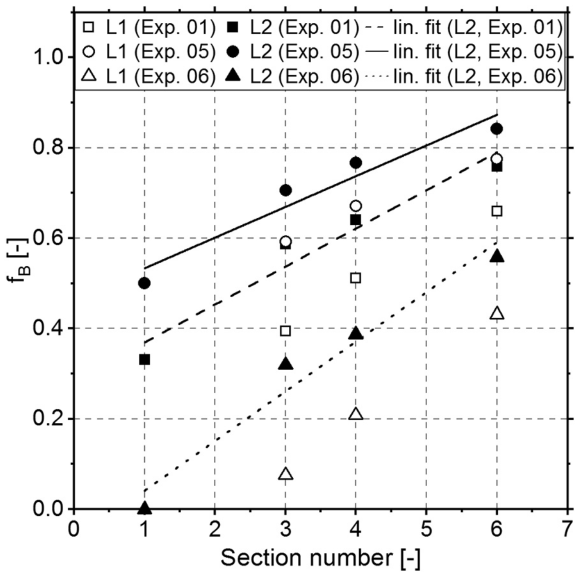

Fig. 7. Evaluation of back spillage fraction in dependency of the section number for experiments 01, 05 and 06. The linear fits deliver correlation coefficients (R^2) above 0.95 and slopes in the range of 0.07 to 0.12.

increase of the sectional internal length reduces the capacity of the drum in terms of volume or mass flow conducted through the drum. Three experiments (Exp. 01/05/06) lead to back spillage in all equipment configurations. Therefore, these experiments are evaluated graphically in Fig. 7. Linear correlations between section number and f_B are observable (all correlation coefficients above 0.95) regardless the investigated experiment or internal structure. The slopes are in the range of 0.07 to 0.12 for all configurations and thus comparable. With the longer versions (lengths nearly congruent to drum length) even the bare drum (1 section) fits into the linear correlation as presented in Fig. 7. The absolute amount of back-spillage is affected by the experiment-number and thus the set of operational parameters. Furthermore, the length of the sections impacts on the absolute amount of back-spillage, but the impact is not constant considering the different operational states. Additionally, f_B evoked by different internal lengths at constant operational states converges with section number.

As first “rule of thumb” for the design of rotary drums with sectional internals in practice, it can be deduced from these findings, that each additional section reduces the capacity of the drum by approximately 12%. This is a conservative estimation, since the impact is lower with shorter length of sectional internals.

To complete the relationship between feed mass flow and residence time, the experimental data of hold-up is taken into account and the hydrodynamic residence times are calculated via an adjustment of Eq. (3):

$$\tau_{hyd} = \frac{m_{hold}}{\dot{m}_F \cdot (1 - f_B)} \tag{14}$$

The results are presented in Table 6 together with the relative deviation of the hydrodynamic residence time from the experimentally determined MRT.

In order to evaluate the deviations regarding the equipment configuration for the whole DOE, mean values and standard deviations (Std.) have been calculated.

Although some specific configurations show deviations above 10% (e.g. EC3L1), the mean values of relative deviation are all equal or below 10%, which is a good congruence taking into account, that the experimental results used in the calculations always include an error. The overall mean deviation is calculated to 8.4 ± 5.3%.

This finding is a further indication of capacity limits, since it proves that the changes in MRT are evoked by lowering the effective mass flow passing the drum as represented by Eq. (14).

3.1.3. Hold-up

As described in the experimental procedure (2.3) the retained mass is measured after each experimental run as hold-up. The results are presented graphically in Fig. 8.

In analogy to the MRT, the characteristic trends can be found here, too. Sectional internals lead to an increase in hold-up, which is plausible due to the higher residence times and the relationship of hydrodynamic MRT (Eq. (3)).

However, in contrast to the MRT values, the pattern of the hold-up masses is not as clear. The reason for this is the back spillage, explained in 3.1.2.

For most configurations, the capacity limit concerning solid hold up seems to be reached. A higher mass flow / volume flow cannot be transported through the drum. Especially for the experiments with low rotational frequency and low inclination angle, (cf. Exp. 01/05 for all configurations) the hold-up is almost independent from the feed mass flow rate and also from the number of sections. Even an increase of the internal length does not lead to higher hold-ups here. The capacity limits seem to be reached already. In contrast to this finding, however, the MRT is increasing further. This can be explained by the increase in back-spillage, reducing the effective mass flow passing the drum. This is what causes a higher MRT for a similar hold-up in the drum.

Therefore, it can be concluded that rotary drums with sectional internals have a specific capacity limit concerning the retained volume or hold-up mass. Overfilling beyond this limit is not possible, a further increase of the feed volume flow simply results in an increase in back spillage flow.

Table 6
Calculated hydrodynamic residence times [s] with relative deviation from the measured MRT.

	τ_{hyd} [s]							rel. Deviation [%]						
	L1				L2			L1				L2		
Exp.	EC1	EC2	EC3	EC4	EC2	EC3	EC4	EC1	EC2	EC3	EC4	EC2	EC3	EC4
1	385	559	746	1156	845	1041	1581	20.4	11.3	11.5	14.7	14.2	10.8	15.2
2	159	234	288	433	316	409	567	8.8	9.1	9.3	12.9	12.0	11.1	11.9
3	158	248	305	369	300	388	585	8.2	22.8	27.8	9.6	10.5	9.5	12.1
4	59	81	93	127	105	139	185	1.1	2.5	3.4	3.9	2.9	7.4	3.5
5	329	534	710	1102	757	1014	1521	5.8	9.0	7.6	11.1	2.9	9.3	15.9
6	134	207	266	409	298	355	530	3.9	6.1	7.3	9.6	11.5	−2.3	7.5
7	149	222	267	407	297	397	581	3.3	6.5	5.1	8.5	5.5	7.7	10.9
8	57	79	91	127	102	133	189	0.6	0.8	0.9	4.1	1.9	3.1	4.0
9	113	176	200	282	236	281	410	6.5	8.6	7.1	8.2	13.3	7.5	11.0
							Mean	6.5	8.5	8.9	9.2	8.3	7.1	10.2
							Std	5.9	6.3	7.7	3.6	4.9	4.3	4.4

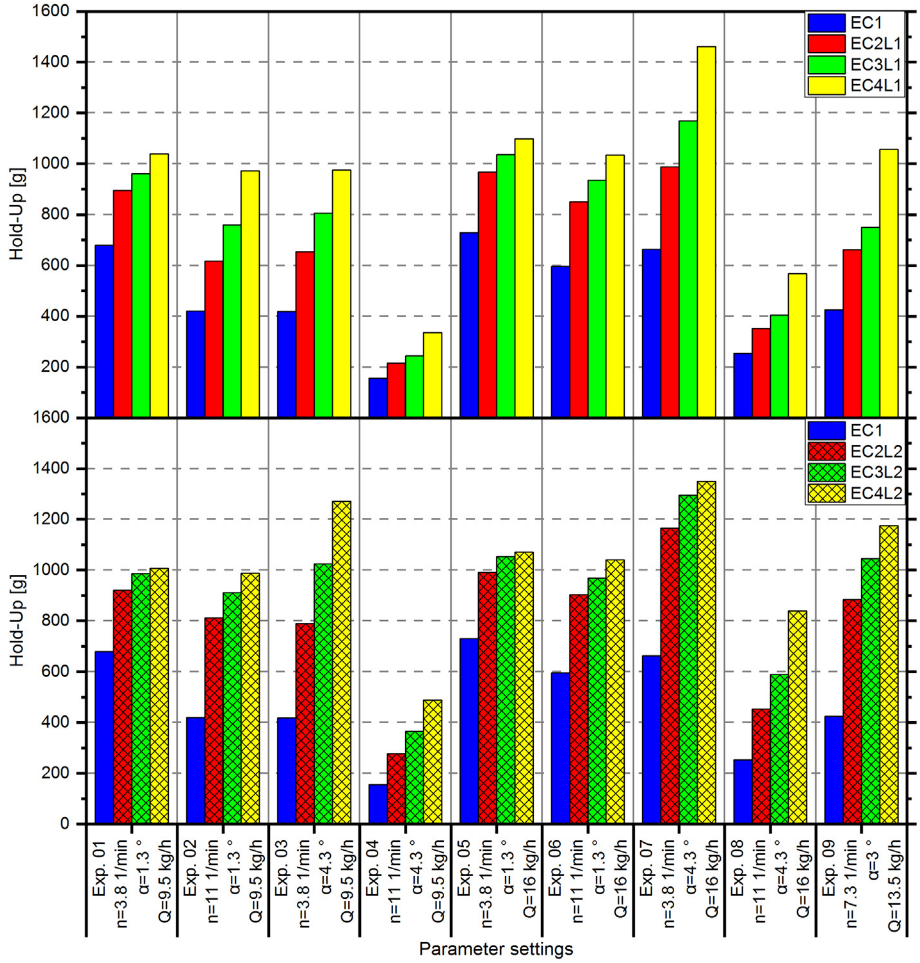

Fig. 8. Diagram of all measured hold-up values for all experiments. Upper diagram: short sectional internals (L1); lower diagram: long sectional internals (L2).

3.1.4. Axial dispersion-coefficients (D_{ax})

Besides residence time, the explained method (chapter 2.5) is used to provide information on the residence time distribution and therefore on the axial mixing in the drum. Thus, the axial dispersion coefficients (D_{ax}) were evaluated from the measured data via the variance of the Gaussian distribution. They provide an impression on the impact of the investigated sectional internals on the axial mixing in the drum. The values are presented in Fig. 9.

First of all Fig. 9 shows that all axial dispersion coefficients are in the range of 10^{-4} to 10^{-7} m^2/s (range of the diagrams). This is exactly the range of measured values for bare drums delivered in the literature by Sherritt et al. [32] and also found in earlier measurements [27,32]. Except a few outliers, the values are very narrow to 1×10^{-6} m^2/s.

The data indicates that axial mixing is reduced by sectional internals with increasing section number, particularly for the short internal configurations. The reason for this is an estimated change in mixing mechanism as described for the cross section in [28]. Mixing in the bare drum takes place in the active layer, which is not that distinct when sectional internals are used. The active layer is dependent on the operational parameters, particularly the rotational speed, evoking the different values of D_{ax} for different operational states. Focusing on the lower diagram with the longer internal configuration the dispersion coefficients are nearly constant to a value of 10^{-6} m^2/s regardless the operational parameters. This supports the approach of changed solid movement.

Comparing the changes caused by the sectional internals with the variation due to parameter settings, their impact is comparable. Since changes of operational parameters are usual in industrial practice, the influence of sectional internals on the axial mixing is considered as negligible.

3.1.5. Effects from DOE

The statistical design of the experimental runs enables the analysis of the effects or sensitivities on the measured values created by the parameter changes. Effects or sensitivities are the mean differences in a measured quantity introduced by the investigated parameter changes. The results are shown in Fig. 10. All diagrams are structured similarly and the bare drum (EC1) is the central value because of its benchmark character. From the center to the outer bars, the number of sections is increasing, grouping the shorter versions (L1) above the bare drum (EC1) and the longer versions (L2) with hatchings below EC1. The bar colors are equal to the other diagrams.

The effects on MRT (Fig. 10, top) prove the finding, that mass flow is not affecting the MRT at all. This is valid for all drum configurations in the investigated parameter space. Further effects of the other operational parameters are negative. This is plausible because increasing the inclination or rotational speed of the drum leads to lower MRT. The effects evoked by drum inclination compared to rotational speed of the drum are nearly equal considering each configuration. Comparing the different equipment configurations, the increasing number of sections leads to higher impact of parameter changes. That means higher sensitivities of the system. The trends are equal regardless to the length of the internals. However, the absolute effects are significantly higher with the longer internal version. Thus, the system sensitivity is dependent on the internal lengths. Sectional internals are therefore ideal in

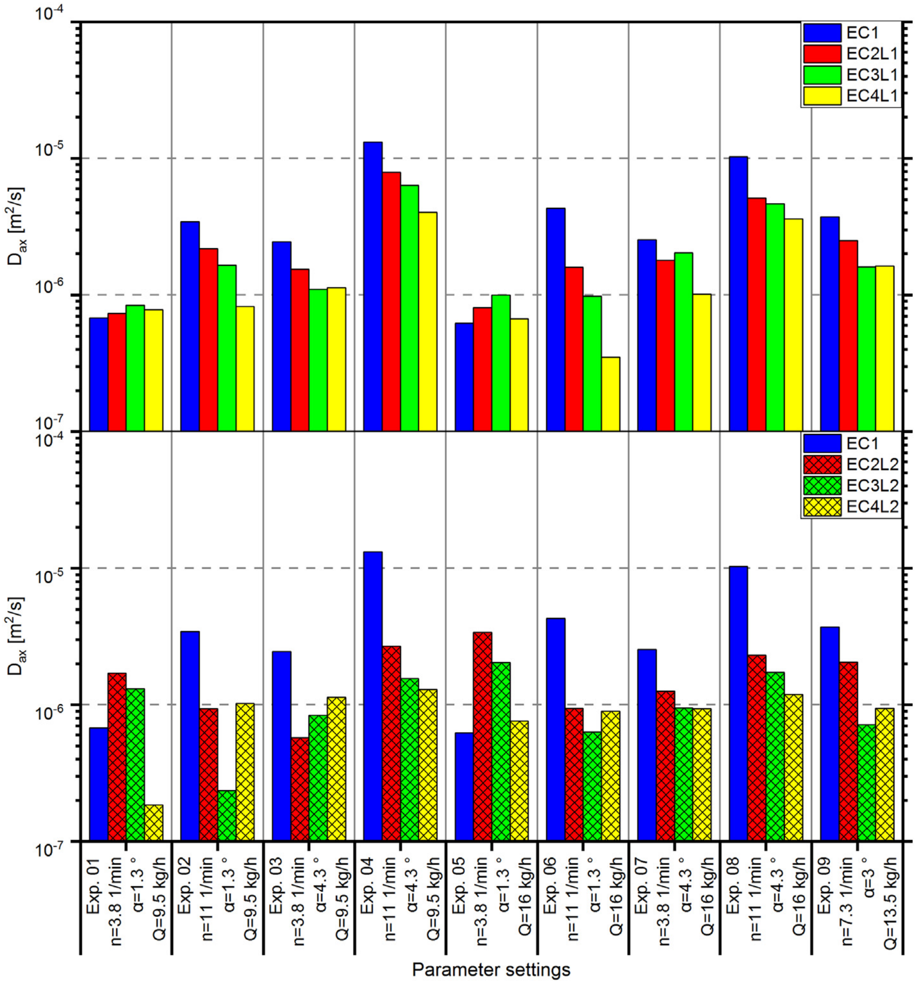

Fig. 9. Diagram of all evaluated axial dispersion coefficients. Upper diagram: short internals (L1); lower diagram: long internals (L2).

applications where e.g. the residence time has to be adjusted very precisely (e.g. production of catalysts [38] etc.).

Considering the effects on the drums hold-up now (Fig. 10, middle), impact of the mass flow on the hold-up is observable in contrast to MRT. A higher feed mass flow leads to higher hold-ups in the drum. Keeping in mind that there is no impact on MRT automatically evokes the need for higher back-spillage reducing the effective mass or volume flow passing the drum. Sectional internals slightly emphasize the impact of mass flow changes, but there is no clear trend regarding section number. Changes in drum inclination affect the hold-up in the drum, but sectional internals (short version) have no significant additional impact compared to the bare drum. The sensitivity is comparable to the bare drum in its magnitude. The evaluation of long sectional internals produced lower impact with increasing section number. With high section numbers (EC4L2) the influence of inclination on hold-up becomes more and more negligible. Keeping the back-spillage results in mind, this is also an indicator for capacity limits. The highest impact on hold-up is generated by the drums rotary speed in the investigated parameter space. Comparing the different equipment configurations, a different pattern to MRT-effects is observable. The rotary speed effects with sectional internals are higher in comparison to the bare drum, but in contrast to MRT-effects, the section number has no further significant impact. Furthermore, the length is not additionally influencing the effect. Again, the explanation to this finding is back spillage.

The last diagram in Fig. 10 (bottom) shows the effects on the axial dispersion coefficient or axial mixing. Again, the mass flow is not strongly influencing the axial mixing. The impact of parameter changes in inclination and rotational speed on axial mixing is comparable. Additional sections decrease the sensitivity in the shorter version while the effects for the long internal version are much smaller (one magnitude). This finding fits to the evaluations from section 3.1.4, where nearly constant dispersion coefficients have been found and is undermined by the switch in particle movement, no longer comparable to bare drums as found in [28]. Thus, with sectional internals the axial mixing becomes more and more independent from operational parameters with increasing section number or length of internals. Because the magnitude of dispersion coefficients is rather constant, this also supports the conclusion in 3.1.4, that axial mixing in general is not affected significantly by sectional internals. Since rotary drums have low axial dispersion and are satisfactorily estimated as plug-flow reactor, sectional internals emphasize this characteristic behavior. No further attention has to be paid on axial mixing or broad RTDs with sectional internals.

Further analysis of the DOE is not feasible because the check of midpoint experiments shows that the deviation of linear models is very high. This is plausible considering the models of Chatterjee or Saeman with nonlinear dependencies concerning inclination or rotational speed of the drum.

3.2. Qualitative observations

The full optical accessibility of the drum reveals additional phenomena of the solid bed, which are analyzed and described qualitatively. First, a thin particle layer (estimated monolayer) has been observed in all experiments on the inner drum wall due to electrostatic adhesion

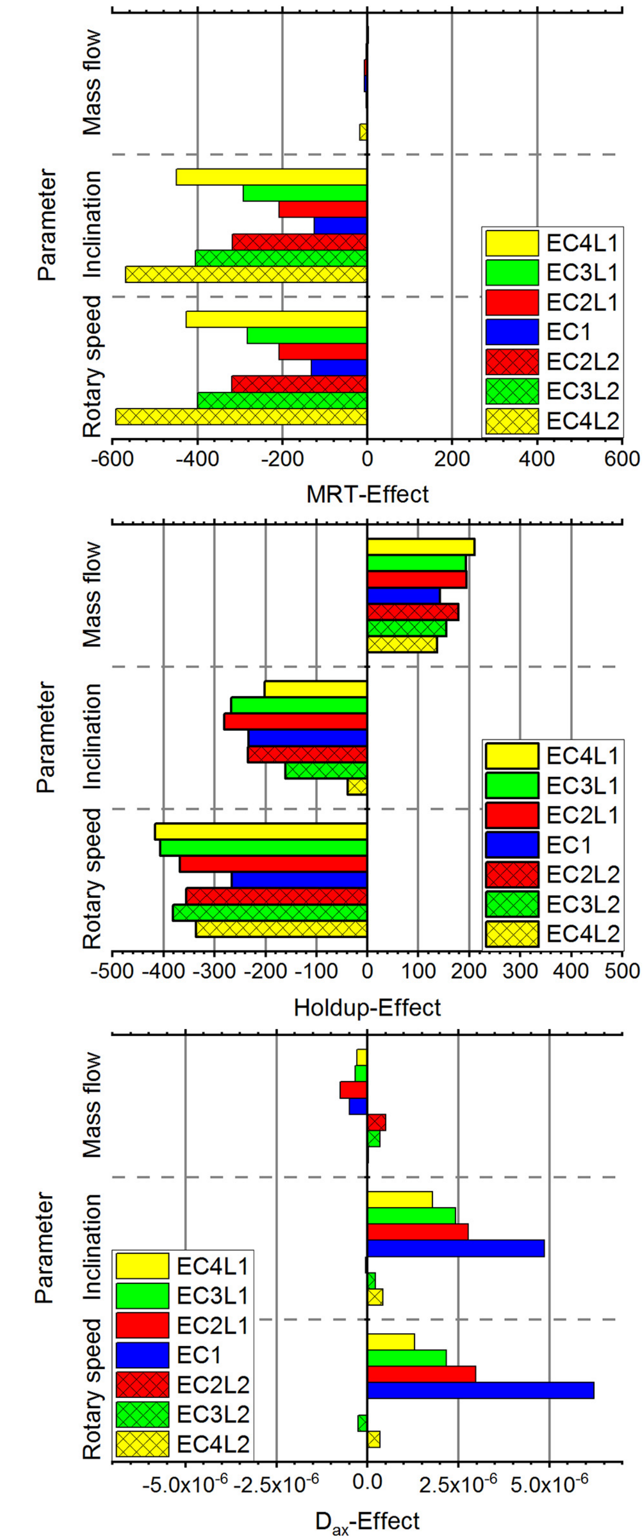

Fig. 10. Effects of parameter changes calculated from DOE on: Top: MRT; Middle: Holdup; bottom: D_{ax}.

of particles. However, a view inside the drum was still possible. Besides that, no further influences by electrostatic charging could be found by visual investigation.

An observation of the drum with a horizontal side view along the drum's axis directly leads to the axial shape of the solid bed. In Fig. 11 two different states are exemplarily presented.

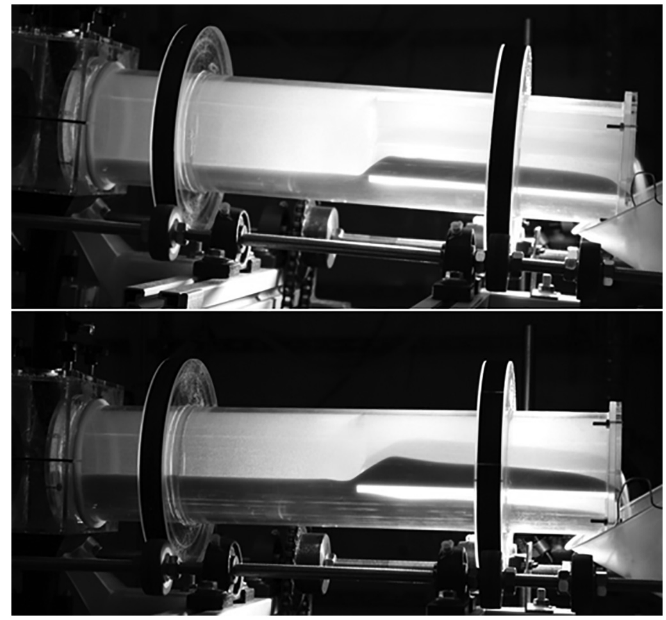

Fig. 11. Exemplary shadowgraph pictures of the side view for the configuration EC3L1 with: 11 rpm; 9.5 kg/h; top: inclination 4.3°; bottom: inclination 1.3°.

The pictures in Fig. 11 have been taken as shadowgraph pictures with a LED illumination from behind the drum to enhance the visibility of the solid bed. The two vertical dark bars are evoked by the treads of the drum's tires. The horizontal dark areas show the shadows of the solid bed in the drum. The bright line below the solid bed in the right part of the drum (outlet end) is the radial front face of one section spoke (PMMA) enlightened by the backlight illumination. Both pictures are made with the same drum configuration (4 sections with a length of 225 mm, EC3L1) and only differ in the drum's inclination. The cross sectional position of the spoke towards the viewer is nearly horizontal in both pictures.

As can be seen clearly from the measured data (section 3.1), the inclination has a massive impact on hold-up and residence time. This can also be observed qualitatively in the heights of the solid beds. The solid bed height in the lower picture with a lower inclination is significantly higher, especially at the beginning of the sectional internal part (transition region between bare drum and sectional internals). The bed height in the bare drum part is additionally clearly higher in the lower picture.

Furthermore, looking from the outlet end of the drum, the characteristic solid bed profile predicted by the Saeman-model for the bare drum can be recognized, although the model originally was developed for bare drums. An exception to this finding is the transition region from sections to bare drum. Due to the filling of the segments, a negative slope of the solid bed height is created, since a part of the solid material is flowing backwards, when the section is lifting above the solid bed of the bare drum.

In the bare drum part (inlet end), the solid bed height is nearly constant. This is plausible because the sectional internals increase hold-up and residence time (see experimental data) and therefore the solid bed is retained backwards to the inlet end. This behavior for retained solid beds in rotary drums is also known from investigations in rotary drums with constriction dams by Kramers et al. and Scott et al. [14,18].

The qualitative observation leads to the conclusion that a mathematical modeling of the solid transport in rotary drums with sectional internals with a modified Saeman-model should be feasible.

4. Conclusion and outlook

This novel study presents the experimental investigation of the impact of sectional internals on the rotary drum process behavior. The examination focuses on the number of sections and the length of the sectional internals. The following key findings regarding the effects of sectional internals can be summarized:

- In general, sectional internals have a significant impact on the solids transport in rotary drums.
- The dependencies of the process behavior are similar to bare drums concerning operational parameters as drum inclination, rotary speed and feed mass flow. However, the absolute values for MRT etc. and the differences of these values are larger with sectional internals resulting in a higher sensitivity of the system.
- The MRT as well as the rotary drum hold-up are increasing with the number of sections and for constant section numbers with their lengths up to a factor of 4 (for MRT). A model approach based on increase factors to describe this behavior has been provided here.
- The capacity of the drum regarding mass flow decreases antiproportionally to the number of sectional internals and their lengths.
- If the capacity limit is reached, further increase of feed mass flow only leads to increase of back spillage and therefore reduces the effective mass flow passing the drum.
- The impact of sectional internals on the axial dispersion coefficient (D_{ax}) and therefore on axial mixing is negligible. A slight trend towards lower dispersion coefficients is observable. Values for D_{ax} are between 10^{-4} to 10^{-7} m^2/s.
- The bed height profile proposed by the Saeman-model can be observed, even when sectional internals are installed.

The findings on the process behavior of rotary drums with sectional internals can be used as a major step and improvement in process optimization and design of industrial rotary drums. The knowledge of increased sensitivity provides the use of sectional internals for the production of high tech materials, such as e.g. pigments [5] or catalysts. These products need very precise adjustment in the production processes. In contrast to that, the decreasing capacity limits due to the found back-spillage have to be considered in designing rotary drums/ kilns with sectional internals.

To our knowledge and also after an intense literature study, the derived correlations have not yet been investigated, except first measurements presented in [27,39]. Therefore, the described findings are an important step in the further improvement of rotary drum equipment and rotary kilns in terms of operating mode and efficiency, saving investment and operating costs, reducing energy demand and resource consumption and finally the more targeted adjustment of product properties and the assurance of product quality for the customer.

Nevertheless, the results of this study will be used in the one dimensional modeling of rotary drums with sectional internals together with further findings [40,41]. The qualitative observation of the bed height profile will be used to develop a modified Saeman-model to describe the process in further steps. Experiments with other drum dimensions have to be conducted to check the validity of scale-up for the findings and for the model approaches. Finally, the results will be introduced in modeling of industrial rotary drum processes. The model validity and therefore the validity of the findings in this study will be checked with real industrial processes on this way.

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Notation

Nomenclature

c	$[g \cdot g^{-1}]$, Concentration
C	$[-]$, Concentration (dimensionless)
D_{ax}	$[m^2 \cdot s^{-1}]$, Axial dispersion coefficient
D_x	$[m^2 \cdot s^{-1}]$, Dispersion coefficient (x direction)
D	[m], Diameter
E	$[s^{-1}]$, Residence time distribution (density distribution)
E_ϑ	$[-]$, Residence time distribution (dimensionless)
f_B	$[-]$, Back-spillage fraction
h	[m], Solid bed height
$\dot{V}$	$[m^3 \cdot s^{-1}]$, Volumetric feed rate
V_{hold}	$[m^3]$, Hold-up volume
L	[m], Length of the drum
$\dot{m}$	$[kg \cdot s^{-1}]$, Mass flow
$\dot{m}_F$	$[kg \cdot s^{-1}]$, Feed mass flow rate
m_{hold}	[kg], Hold-up mass
n	$[min^{-1}]$, Rotational speed/frequency
Pe	$[-]$, Peclet number
R	[m], Radius of the drum
t	[s], Time
$\bar{t}$	[s], Mean residence time
τ_{hyd}	[s], Hydrodynamic residence time
v_x	$[m \cdot s^{-1}]$, Velocity (x direction)
z	$[-]$, Length (dimensionless)
β	[°], Inclination of the drum
ϑ	$[-]$, Dimensionless time
θ	[°], Angle of repose

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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